

Lexical:

A word is used to describe something in the requirement but in the given context could have multiple meanings (i.e., Program, a code program or a program of study)

Syntactic:

If a sentence forks or has a word / adjective that applies to multiple other parts (i.e., monitoring log-in attempts and reporting discrepancies.)

Semantic:

The sentence does not show syntactic or lexical ambiguity however there are multiple interpretations. The two separate meanings are usually caused by the sentence requiring further context found outside of the sentence.

(I.e., "The system shall implement a mechanism to encrypt and decrypt electronic protected health information (ePHI).")

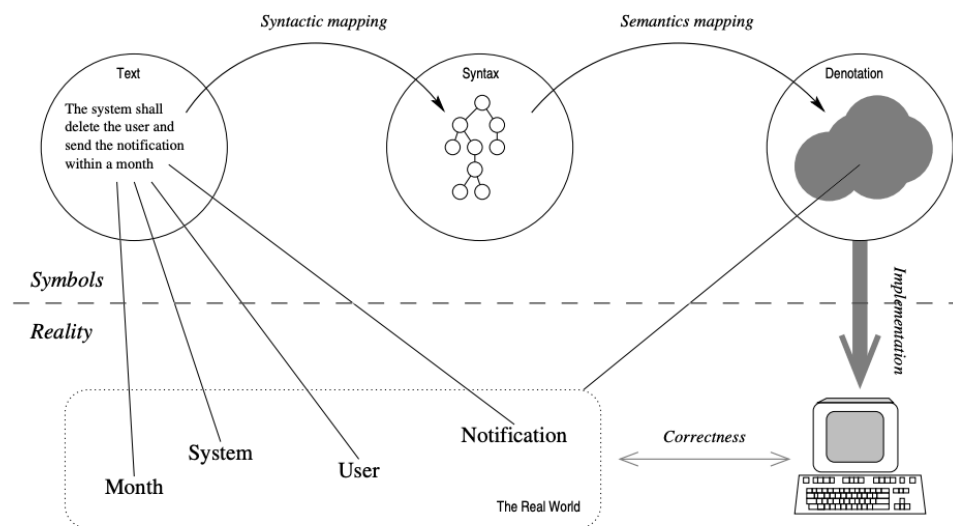


Fig. 1. The theoretical framework for the occurrence of ambiguity.

Vagueness:

The sentence includes an adjective that is correct, however it should require more specific details that provide clear instructions for completion and remove borderline cases or relativity.

(I.e., The system shall terminate a user sessions after an appropriate period of inactivity)

(Should be more specific, appropriate itself is fuzzily defined)

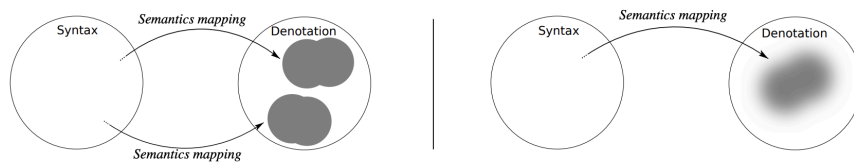


Fig. 3. A visual rendition of the difference between ambiguity (left) and vagueness (right).

Incompleteness:

When something is an action noun i.e., (emergency), and it is not described when during this action the requirement must be fulfilled this is incomplete. (e.g., The system shall terminate a user session after a predetermined period of inactivity)

Referential:

The sentence includes incorrect grammar or otherwise makes it unclear for what a noun or pronoun references (i.e., "The system shall allow a personal representative the same level of access as a person over their PHI.")

Source of figures: Gervasi, Vincenzo, et al. "Ambiguity in requirements engineering: Towards a unifying framework." *From Software Engineering to Formal Methods and Tools, and Back: Essays Dedicated to Stefania Gnesi on the Occasion of Her 65th Birthday*. Cham: Springer International Publishing, 2019. 191-210.